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Introduction: At the pre-specified interim analysis of the CameL phase 3 study, camrelizumab plus chemo (carboplatin + pemetrexed) as first-line therapy significantly improved the PFS compared with chemo alone and showed acceptable safety profile in patients with advanced non-squamous NSCLC without sensitizing *EGFR* and *ALK* alterations (2019 WCLC OA04.03). Herein, we provide an update on OS and PFS2 (time to second progression) based on long-term follow-up. **Methods:** Eligible patients were randomized 1:1 to receive 4 to 6 cycles of chemo with (n=205) or without (n=207) camrelizumab, followed by pemetrexed with or without camrelizumab as maintenance therapy. Crossover to camrelizumab monotherapy was permitted for patients in the chemo alone group who had radiological disease progression. Data on post-study anticancer therapy and outcomes were collected. PFS2 was defined as time from randomization to disease progression after the first disease progression event or death, whichever occurred first. There was no multiplicity adjustment, and nominal one-sided P values are presented. ClinicalTrials.gov number: NCT03134872. **Results:** At data cutoff on February 25, 2020, the median follow-up duration was 19.3 months (IQR 9.8−23.7). 35 (17.1%) patients in the camrelizumab plus chemo group and 18 (8.7%) in the chemo alone group were still receiving the assigned first-line study treatment. Camrelizumab plus chemo prolonged median overall survival, as compared with chemo alone (27.9 months [95% CI 21.9−not reached] vs 20.5 months [95% CI 15.9−24.4]; HR 0.73 [95% CI 0.55−0.96]; P=0.0117). Second-line or later therapy was received by 98 (47.8%) patients in the camrelizumab plus chemo group and 135 (65.2%) patients in the chemo alone group. Median PFS2 was 18.9 months (95% CI 15.7−21.2) vs 12.5 months (95% CI 10.6−15.6) in patients who received vs did not receive first-line camrelizumab (HR 0.66 [95% CI 0.52−0.84]; P=0.0004). Benefits in OS and PFS2 with camrelizumab plus chemo were also found in patients with PD-L1 TPS ≥1% (Table). The safety profile was consistent with the previous report at interim analysis.

Conclusion: Camrelizumab plus carboplatin and pemetrexed as first-line therapy showed long-term benefit in OS and PFS2 compared with chemo alone in Chinese patients with advanced non-squamous NSCLC without *EGFR* and *ALK* alterations, despite 48.3% of patients in the chemo alone group receiving subsequent immunotherapy. No new safety signal was observed. This combination represents a potential standard first-line therapy for these patients. **Keywords:** Camrelizumab, NSCLC, PD-1

	OS				PFS2			
	No. events/No. patients				No. events/No. patients			
	Camrelizumab plus chemo	Chemo alone	HR (95% CI)	P value	Camrelizumab plus chemo	Chemo alone	HR (95% CI)	P value
All patients (n=412)	44.9%	54.6%	0.73 (0.55 −0.96)	0.0117	60.0%	71.5%	0.66 (0.52 −0.84)	0.0004
Patients with PD-L1 TPS ≥1% (n=255)	38.4%	49.6%	0.70 (0.48 −1.02)	0.0318	53.6%	67.5%	0.64 (0.46 −0.88)	0.0027

P79.03

A Phase 3 Study of the PD-1 Inhibitor Retifanlimab (INCMGA00012) Plus Platinum-Based Chemotherapy in 1L mNSCLC: POD1UM-304

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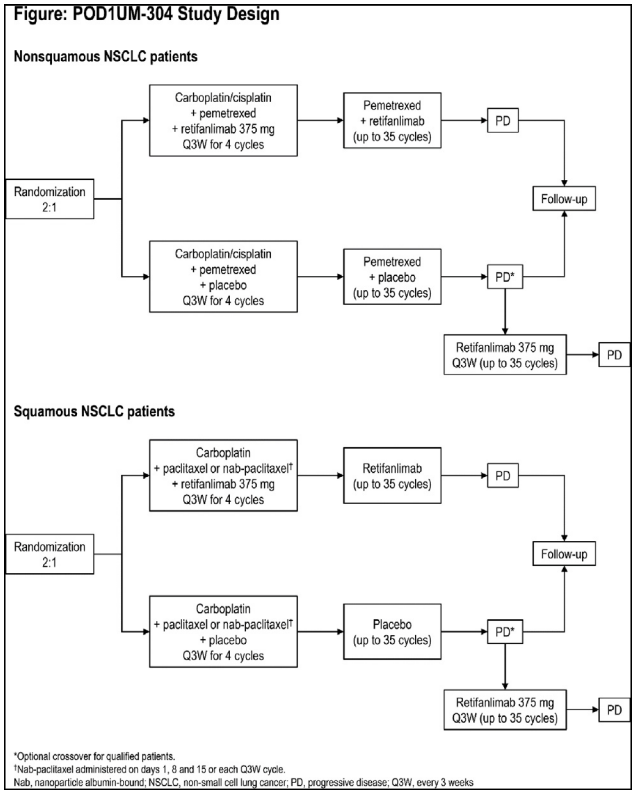
Introduction: Checkpoint inhibitors have significantly improved outcomes in patients with metastatic non-small cell lung cancer (mNSCLC), and are approved as first-line (1L) treatment as monotherapy as well as in combination with platinum-based chemotherapy. Retifanlimab is a an investigational humanized monoclonal anti−programmed death-1 (PD-1) antibody; in the POD1UM-101 phase 1 study (NCT03059823), retifanlimab demonstrated clinical activity and safety profile similar to other currently approved anti PD-1 antibodies in patients with advanced stage NSCLC. POD1UM-304 (NCT04205812) is a randomized, double-blind, phase 3 global study designed to investigate the efficacy and safety of retifanlimab plus platinum-based chemotherapy for 1L treatment of metastatic NSCLC. **Methods:** Previously untreated adults ≥18 years of age, with histologically or cytologically confirmed metastatic squamous or nonsquamous NSCLC, and documented absence of driver mutations or gene arrangements for *EGFR*, *ALK*, *BRAF*, or *ROS1* (nonsquamous tumor histology) are eligible. Patients must have measurable disease per Response Evaluation Criteria in Solid Tumors v1.1, an ECOG performance status ≤1, adequate organ function, and be able to provide tumor tissue during screening or have a fresh tumor biopsy after diagnosis of metastatic disease. Neoadjuvant or adjuvant therapy that did not include PD-1/programmed death-ligand 1 (PD-L1) −directed agents is permitted if completed ≥12 months before diagnosis of metastatic disease. Approximately 530 participants will be randomized 2:1 to retifanlimab 375 mg plus platinum-based chemotherapy or placebo plus platinum-based chemotherapy every 3 weeks (Q3W; figure). Participants in the placebo arm will have the option to crossover to retifanlimab monotherapy following disease progression.

Primary study objectives include overall survival and progression-free survival in participants treated with retifanlimab plus chemotherapy versus placebo plus chemotherapy. Secondary objectives include objective response rate, duration of response, and safety and tolerability in the two treatment arms, as well as the pharmacokinetics of retifanlimab 375 mg Q3W when administered in combination with chemotherapy.



Keywords: Phase II and III Clinical Trials, Immune checkpoint inhibitors, Randomized Controlled Trials

P79.04

A Phase 2 Trial of Nivolumab and Temozolomide in Extensive Stage Small Cell Lung Cancer: Interim Efficacy Analysis.

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Introduction: Effective treatment options are limited in patients with extensive stage small cell lung cancer (ES-SCLC) after progression on first line therapy. Temozolomide is active in ES-SCLC, and has been shown to have immunomodulatory impact on lymphoid cells in patients with advanced cancers. Nivolumab is approved as third line therapy. Here we report the interim efficacy analysis of the ES-SCLC cohort of our phase 2 trial of combination nivolumab and temozolomide in patients with advanced small cell lung cancer as second or third line therapy. **Methods:** NCT03728361 is a non-randomized, two-cohort, single-institution, open-label phase 2 study of nivolumab and temozolomide in patients with resistant or refractory ES-SCLC as well as a separate cohort for neuroendocrine tumors. Eligible patients must have progressed after treatment with first line chemotherapy-immunotherapy. Study treatment consists of nivolumab 480 mg IV every 4 weeks and temozolomide 150 mg/m2 for 5 days out of a 28 day cycle (amended from 200 mg/m2). The primary endpoint is best overall response rate (ORR) by RECIST v1.1. For the SCLC cohort, a protocol defined interim analysis would be performed after 15 patients, where $\geq$ 2 objective responses would be considered sufficient to continue accrual to a total of 25 patients. Progression free survival (PFS) and overall survival (OS) were assessed by the method of Kaplan—Meier. Adverse events were graded using CTCAE v5. Here we will present the interim efficacy analysis after 15 patients were accrued in the ES-SCLC cohort. **Results:** Twelve of 15 patients enrolled in the ES-SCLC cohort were evaluable for response at the time of this submission (2 patients were treated under a prior amendment and had not received first line chemotherapy-immunotherapy and were therefore not eligible for primary endpoint analysis and 1 patient did not have imaging as of this submission). Interim efficacy data will be presented for this cohort at the meeting including ORR, PFS, and OS. **Conclusion:** Data for combination nivolumab and temozolomide as second or third line treatment for ES-SCLC after first line chemo-immunotherapy will be presented. **Keywords:** small cell lung cancer, Immunotherapy, Immune checkpoint inhibitor

P79.05

Phase II Randomized Trial of Neoadjuvant Pembrolizumab +/- Chemotherapy for Operable Stage IA3-IIA Non-Small Cell Lung Cancer

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Introduction: Surgery remains a cornerstone of cure for early stage non-small cell lung cancer (NSCLC). The value of systemic therapy for node negative early stage lung cancer is unclear, yet local and distant recurrence rates remain high after surgery. The advent of immunotherapy for advanced NSCLC has created new opportunities for patients. With better side effect profiles, improved outcomes and prolonged survival, immune checkpoint inhibition (ICI) as a single agent or in combination has become a first-line therapy in advanced NSCLC. Emerging data indicate that these advantages may be translated to early stage NSCLC. Previous studies have demonstrated safety and tolerability of neoadjuvant ICI for patients with operable NSCLC, as well as unprecedented major pathological response (MPR) rates, ranging from 20 to 75%. However, the ideal timing and regimen remain unknown. It is also unknown how the degree of pathological response should impact the extent of surgical resection. Determining the likelihood of complete pathological response (pCR) prior to surgery through clinically available biomarkers is therefore key. Radiomic and ctDNA technologies are attractive for this purpose. This phase II study aims to evaluate the use of clinically available biomarkers during neoadjuvant therapy with pembrolizumab alone or in combination with chemotherapy for stage IA3-IIA operable NSCLC to predict the likelihood of